FRC JAVA CODING LAB 7.0

TIÊU CHUẨN PACKAGE OBSERVATION

Quy tắc cho model, evaluator, dependency và migration

Document	Version	Status	Language
OC-02	1.0	FROZEN	Tiếng Việt

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1. Trách nhiệm package

frc.robot.observation contains immutable observation models and pure interpretation helpers. Mechanism-specific types belong in frc.robot.observation.<mechanism>. Cross-mechanism aggregation belongs in a clearly reviewed observation subpackage only when it is genuinely observation, not command coordination.

2. Nội dung được phép

<Mechanism>Observation records or immutable final classes; observation-only enums and vendor-neutral value objects; <Mechanism>ObservationEvaluator pure helpers; concise JavaDoc defining source, units, timing, and validity.

3. Nội dung bị cấm

Vendor API imports; motor controllers and sensors; IO interfaces or IOInputs definitions; NetworkTables, Glass, DataLog, or publisher code; WPILib Command or CommandScheduler; RobotContainer; Xbox or DriverStation reads; mutable caches; timers; control setpoints; safety interlocks; command decisions; subsystem lifecycle methods.

4. Evaluator Hợp đồng

Evaluator are pure, stateless, deterministic, and side-effect free. Their output depends only on explicit parameters. They do not read clocks, environment, singletons, hardware, NetworkTables, DriverStation, or mutable global state.

Use static methods in a non-instantiable final class or an immutable callable object only when configuration is passed explicitly and remains immutable. The same inputs and configuration must produce equal outputs.

5. Quy tắc dependency

subsystem/estimator -> observation is allowed for construction. telemetry -> observation is allowed for consumption. observation -> telemetry, io, subsystem, commands, controls, RobotContainer, hardware, or vendor libraries is forbidden.

A pure evaluator may accept primitive values, immutable Observation inputs, or a deliberately copied vendor-neutral snapshot. It must not accept a live subsystem, IO implementation, publisher, supplier that hides side effects, or mutable IOInputs by retained reference.

6. Trình tự tạo và publish

1. IO implementation updates IOInputs.
2. Subsystem periodic reads the complete snapshot.
3. Subsystem or estimator copies/selects values and invokes any pure evaluator.
4. It creates one immutable Observation.
5. It atomically replaces its latest Observation reference or returns the new value.
6. Telemetry reads the Observation and publishes typed stable topics.
7. Logging may record the same Observation without changing it.

7. Kiểm thử

Model tests cover immutability, equality, units, invalid values, and boundary values. Evaluator tests cover truth tables, thresholds, NaN/infinity policy where applicable, and repeatability. Architecture checks reject forbidden imports and dependency cycles.

8. Ghi chú migration

Do not modify completed lessons. Apply Document C to the next editable lesson by copying the frozen baseline under the normal lesson lifecycle. Inventory current observation types; preserve public meaning; add explicit validity/time only where required; move publication concerns to telemetry; move hardware transport to IOInputs; replace stateful evaluators with pure functions; build and verify after each change.

L10 aligns in package placement, immutable records, subsystem production, and telemetry consumption. Its DriveObservationEvaluator is a useful pattern. A future lesson should formally review whether all mechanism Observations need explicit sample timestamps or validity beyond connected/configuration flags.

9. Kiểm soát thay đổi

Because this package is FROZEN, responsibility moves or breaking model changes require Reason, Scope, Impact, Decision, version update, and migration evidence.

Lịch sử sửa đổi

1.0 | 2026-08-01 | FROZEN | Phát hành ban đầu